

ZHIYUAN WANG

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SUMMARY

Senior undergraduate in Computer Science at the University of Chinese Academy of Sciences (GPA 3.86/4.0), and a Spring 2025 visiting student at UC Berkeley. My research focuses on ML for EDA and data-driven analog circuit sizing, especially surrogate modeling and uncertainty-aware Bayesian optimization. I aim to develop data-efficient machine learning methods that accelerate and improve the reliability of EDA and related scientific engineering workflows.

EDUCATION

University of Chinese Academy of Sciences (UCAS) Sep. 2022 – Jun. 2026 (expected)
B.S. in Computer Science and Technology *Beijing, China*
GPA: 3.86/4.0

University of California, Berkeley Jan. 2025 – May 2025
GLOBE Visiting Program, College of Engineering *Berkeley, CA*

EXPERIENCE

Ningbo Institute of Digital Twin; Eastern Institute of Technology (EIT) Ningbo, China
Research Intern (Advisor: Lei He, Professor Emeritus, ECE, UCLA) Jun. 2025 – Present

- Built a surrogate-modeling pipeline to replace costly simulator-in-the-loop evaluations in SAR-ADC sizing, enabling Bayesian optimization to reach multi-spec design targets with substantially fewer simulation calls.
- Designed an offline, dataset-driven framework with uncertainty-awareness so the BO loop can adaptively choose between surrogate and circuit simulator, improving sample efficiency and robustness under distribution shift.
- Explored Bayesian neural network (BNN)-based optimization and OOD-aware training strategies; second author on an ISCAS submission on SAR-ADC surrogate modeling (under review).

State Key Laboratory of Processors; ICT, Chinese Academy of Sciences Beijing, China
Research Intern (Advisor: Yunji Chen, Deputy Director and Professor, ICT, CAS) Mar. 2025 – Aug. 2025

- Worked on an AI-assisted processor IP library that uses machine learning to automate digital circuit design, focusing on reducing Binary Decision Diagram (BDD) size to improve performance, power, and area (PPA).
- Designed variable-reordering schemes that combine classical heuristics with evolutionary and Bayesian search, cutting BDD node counts and simplifying implementations on standard benchmarks.

Microprocessor Research Center; ICT, Chinese Academy of Sciences Beijing, China
Research Intern (Advisor: Fuxin Zhang, Professor) Jul. 2024 – Aug. 2024

- Upgraded the AOSP compiler toolchain for LoongArch to newer Clang/LLVM and Bionic, enabling reliable Android target builds on the new architecture.
- Built a reproducible cross-compilation environment, documented build configurations and debugging workflows, and resolved ABI, linker, and runtime issues to stabilize the system stack.

Operating Systems Course Project; UCAS Beijing, China
Student Project — C-based RISC-V OS Kernel Sep. 2024 – Dec. 2024

Implemented an OS kernel for RISC-V with over 5,000 lines of code, including process management, virtual memory, a minimal file system, and dual-core scheduling, validated through systematic debugging and stress testing.

Computer Architecture Course Project; UCAS Beijing, China
Student Project — LoongArch32 Pipelined Processor Design Sep. 2024 – Dec. 2024

Designed and implemented a LoongArch32 pipelined CPU (67-instruction subset) with exception/interrupt handling, TLB-based MMU, AXI bus, and instruction/data caches, and brought up the design on FPGA using representative workloads.

HONORS & LEADERSHIP

China Postgraduate IC Innovation Competition: EDA Elite Challenge Contest, Second Prize Nov. 2025
University-level Merit Student Award, University of Chinese Academy of Sciences 2023, 2024, 2025
Undergraduate “Role Model of the Year” Award, UCAS Dec. 2024
University-level Outstanding Student Leader Award, UCAS Jun. 2024

Student Union, UCAS: Vice President (2024 – 2025); Director, External Relations Dept. (2023 – 2024)

SKILLS & RESEARCH INTERESTS

Programming: C/C++, Python, Verilog — **ML/Optimization**: Deep learning (PyTorch), Bayesian optimization, genetic algorithms — **Hardware/EDA**: Computer architecture, analog/mixed-signal circuits, EDA